

ABDULBASIT OLANREWAJU GBOLAHAN

AI/ML ENGINEER

Lagos, Nigeria | [LinkedIn](#) | [GitHub](#) | [Email Address](#)

SUMMARY

High-impact AI Engineer and active AWS Community Builder specializing in large language model (LLM) orchestration, agentic system design, and AI inference optimization. Proven track record of developing multi-agent ReAct loops, deploying scalable Retrieval-Augmented Generation (RAG) pipelines, and managing vector database infrastructure. Adept at bridging the gap between machine learning models and production environments to deliver autonomous, cost-efficient, and highly observable AI applications using LangGraph, Python, FastAPI, and AWS.

TECHNICAL SKILLS

- **Agentic Systems & Orchestration:** LangGraph, LangChain, AutoGen, CrewAI, ReAct Loops, RAG Pipelines, AI Tool-Calling
- **AI Inference & Cost Optimization:** Model Quantization (8-bit), Semantic Caching (Redis), LLMLingua, Context Window Optimization
- **XOps & Observability:** LangSmith, Arize Phoenix, Galileo, Token/Latency Monitoring, Hallucination Guardrails
- **Vector DBs & Search Infrastructure:** Pinecone, Qdrant, pgvector, HNSW Optimization, High-Dimensional Indexing
- **Data Engineering & Context:** Apache Airflow, Apache Kafka, Databricks, Snowflake, Multimodal Data Pipelines, Data Knowledge Graphs
- **Semantic Layer & Logic:** dbt Semantic Layer (MetricFlow), Cube.js, Enterprise Business Metrics as Code
- **Programming & Frameworks:** Python, FastAPI, API Development, RESTful APIs
- **Tools & Methodologies:** Git, Power BI, Technical Documentation, Agile Development, System Architecture

PROFESSIONAL EXPERIENCE

AI Engineer, Arthurite Integrated **Sep 2025 – Aug 2026**

- Built and deployed the "AI News Caster" platform, engineering unstructured multimodal data pipelines (text, audio, video) to process over 100,000 words daily across multiple Nigerian languages.
- Orchestrated real-time streaming ETL pipelines utilizing Apache Kafka and Apache Airflow to feed active metadata and content into localization engines.
- Scaled cloud architectures using AWS S3, implementing semantic caching to reduce redundant API queries and optimize inference costs by 40%.

Data Analyst, Kuagi Resources **Mar 2022 – Jan 2026**

- Implemented a unified semantic layer utilizing dbt to centrally define business metrics as code, ensuring all downstream AI agents and Power BI dashboards pulled from a single mathematical definition without hallucinating logic.
- Streamlined ELT workflows and managed schema evolution, transitioning legacy reporting into AI-ready data ecosystems.
- Delivered actionable data-driven insights to cross-functional teams utilizing Python and SQL, reducing manual reporting time by 45% and accelerating executive decision-making cycles.

Technical Documentation Contributor , Hackmamba

Apr 2025 – Jun 2025

- Authored high-quality technical documentation for open-source developer tools, translating complex system architectures into accessible developer guides.
- Recognized by the community for excellence in clarity and impact, improving open-source contributor onboarding efficiency by 30% and significantly reducing time-to-first-commit.

Machine Learning Engineer, Omdena

May 2023 – Nov 2023

- Engineered and implemented machine learning algorithms for an impactful global project focused on determining environmental water retention capacity.
- Collaborated with a distributed, international team of data scientists to process large-scale datasets, achieving a robust model prediction accuracy of 88% and reducing false positive readings by 20%.

PROJECTS

Lead Developer, ChainQuery-AI App

Jan 2026

- Architected an autonomous AI agent utilizing LangGraph and multi-agent collaboration to seamlessly translate natural language prompts into optimized SQL queries.
- Designed dynamic reasoning and execution (ReAct) loops with AI tool-calling, enabling the agent to securely access external enterprise databases.
- Improved database accessibility for non-technical users, decreasing query generation time by 75% and achieving a strict query execution accuracy rate of 94%.

AI Engineer, AgriSabi Project

Nov 2025 - Present

- Built robust Retrieval-Augmented Generation (RAG) pipelines over domain-specific crop and soil management documents.
- Optimized vector database performance utilizing pgvector, tuning HNSW algorithm parameters to balance search accuracy with sub-millisecond query latency.

EDUCATION

Bachelor of Engineering in Agricultural and Biosystems Engineering

Oct 2018 - Oct 2024

University of Ilorin | Ilorin, Nigeria

- Major in Soil and Water Engineering.

CERTIFICATIONS & AFFILIATIONS

- AWS Certified Cloud Practitioner | Amazon Web Services (Nov 2025 – Nov 2028)
- Kubernetes Cloud native Associate | The Linux Founation (Jul 2026 – Jul 2028)
- Machine Learning Certification | DataTalksClub (Feb 2025)
- Agentic AI Certificate | DivVerse Labs (Apr 2025)
- Developer Relation Certification | DXMentorship (May 2025)
- AWS Community Builder | Amazon Web Services (Active)
- AWS Certified AI Practitioner | (In Progress)